PROPOSITIONS AND THEIR CONVERSES

Propositions

✓ **Concept:**

- A sentence that is **either true or false** (but not both) is called a **proposition** (or a statement).
- A proposition does not have to be true. A false sentence is also a proposition, as long as we can clearly say that it is false.
- Questions, orders, wishes and personal opinions are **not** propositions, as they cannot be called true or false.
- A sentence with an unknown, like 'x + 3 = 7', is also not a proposition until the value of x is given, as it is true for x = 4 and false for other values.
- Many propositions are written in the form **'If X, then Y'**.
  - **X** is the **'if' part**: the **condition** (also called the **hypothesis**) that is given.
  - **Y** is the **'then' part**: the **result** (also called the **conclusion**) that follows from the condition.
- 'If X then Y' is also written as **'X implies Y'**, or in short **X ⇒ Y**. Both are used **interchangeably** (in place of each other); they have the same meaning.
- The same meaning can be given in other words also, such as **'Y when X'**, 'Y whenever X' or 'every X is Y'.
- **Note:** whatever the wording, the **condition stays the same**. In 'Y when X', X is still the condition.

✓ **Examples:**

Sentence	Proposition?
7 is a prime number.	Yes (true)
5 is an even number.	Yes (false)
What is your name? / Close the door.	No (a question / an order)
Mathematics is the most interesting subject.	No (an opinion)
x + 3 = 7	No (depends on the value of x)

- **Example 1: Statement 1:** *If two sides of a triangle are equal, then the angles opposite the equal sides are equal.* This is a true proposition (proved in an earlier grade). Here, X = 'two sides of a triangle are equal' and Y = 'the angles opposite them are equal'.
- **Example 2:** 'If a number is a perfect square, then it has an odd number of factors' and 'A number has an odd number of factors **when** it is a perfect square' have the same meaning. In both, the condition is 'the number is a perfect square'.
- **Example 3:** 'Every multiple of 6 is a multiple of 3' means 'If a number is a multiple of 6, then it is a multiple of 3'.

✓ **Questions:**

Q1. Write the 'if' part and the 'then' part of: 'Two triangles have the same area when they are congruent.'

➢ The condition is 'two triangles are congruent (exactly the same in shape and size)', so the **'if' part** is 'two triangles are congruent'.

➢ The **'then' part** is 'they have the same area'.

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➢ In 'if-then' form: If two triangles are congruent, then they have the same area.

Q2. Write 'Every square is a rectangle' in the 'If X, then Y' form.

➢ If a quadrilateral (a closed figure with four sides) is a square, then it is a rectangle.

Converse of a Proposition

✓ **Concept:**

- When the 'if' part and the 'then' part of a proposition are **interchanged**, the new proposition is called its **converse**.
- The converse of **'if X then Y'** is **'if Y then X'**.
- The converse of the converse is the original proposition. So the two are converses of **each other**.
- If the proposition has a first sentence that only sets up the situation (like 'Let a, b, c be the sidelengths of a triangle'), that sentence is kept **as it is**. Only the 'if' part and the 'then' part are interchanged.
- The converse is a **new statement**. If a proposition is true, its converse is **not always** true; it has to be checked on its own.
- In fact, all four cases are possible:

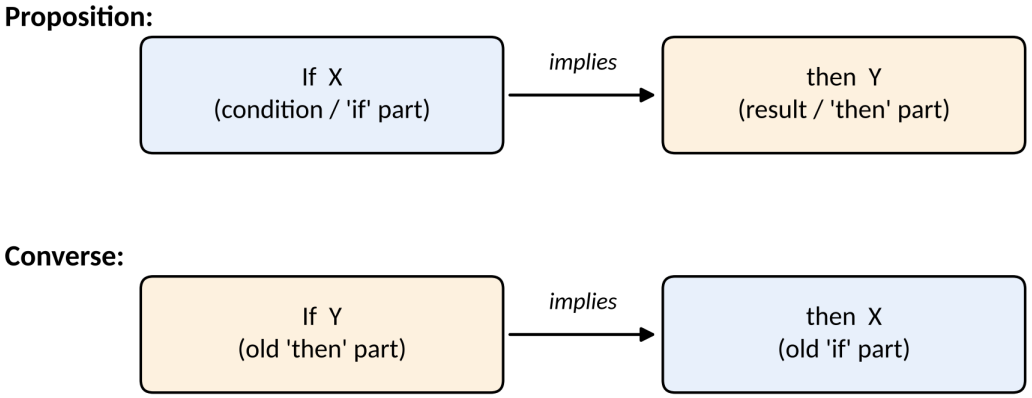


Fig. 1: The 'if' part and the 'then' part of a proposition and its converse

Table 1: The four possible cases

Proposition	Converse	Example
True	True	If n is a perfect square, then it has an odd number of factors.
True	False	If a number is a multiple of 6, then it is a multiple of 3.
False	True	If two triangles have the same area, then they are congruent.
False	False	If a number is a multiple of 2, then it is a multiple of 3.

- Each of these examples is checked in detail in Topics 3 and 4.

✓ **Examples:**

- **Example 1:** the converse of Statement 1 is **Statement 2:** *If two angles of a triangle are equal, then the sides opposite the equal angles have equal lengths.* Here, both statements are true (proof in Q1 below).
- **Example 2: Proposition:** Let a, b, c be the sidelengths of a triangle. If the triangle is right-angled (with

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c as the longest side), then $a^2 + b^2 = c^2$. **Converse:** Let a, b, c be the sidelengths of a triangle. If $a^2 + b^2 = c^2$, then the triangle is right-angled. (The first sentence stays the same.)

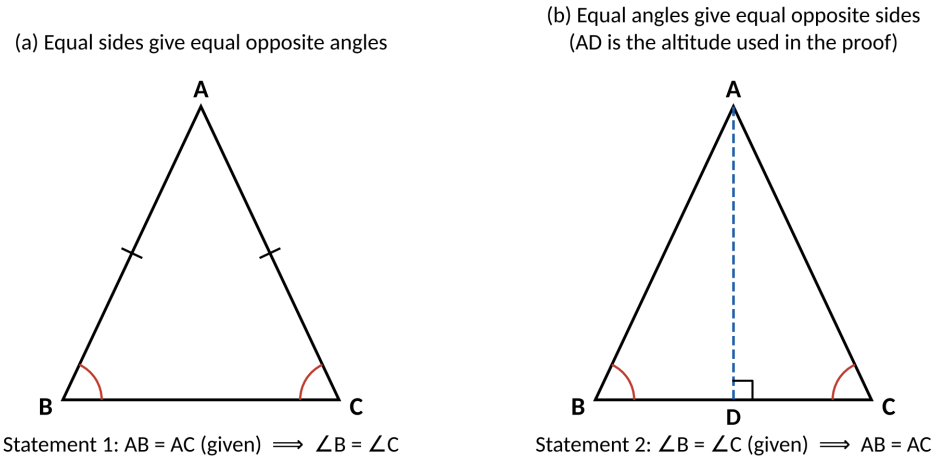


Fig. 2: Statement 1 and its converse, Statement 2, in triangle ABC

✓ **Questions:**

Q1. We have proved Statement 1 in an earlier grade. Is Statement 2 (its converse) true? Prove it. (Hint: draw the altitude from the vertex containing the third angle.)

- Yes, Statement 2 is true.
- Take ΔABC with ∠B = ∠C. Draw the **altitude** (the perpendicular line drawn from a vertex to the opposite side) AD from A to BC, so that ∠ADB = ∠ADC = 90°.
- In ΔABD and ΔACD: ∠B = ∠C (given), ∠ADB = ∠ADC = 90°, and AD = AD (common).
- So ΔABD ≅ ΔACD by the AAS rule. Hence AB = AC (corresponding parts of congruent triangles).

Q2. It can also happen that both a proposition and its converse are false. Give an example.

- **Proposition:** If a number is a multiple of 2, then it is a multiple of 3. **False**: 4 is a multiple of 2 but not of 3.
- **Converse:** If a number is a multiple of 3, then it is a multiple of 2. **False**: 3 is a multiple of 3 but not of 2.

Q3. The shortcut to check divisibility by 3 is to add the digits of the number and check if the sum is a multiple of 3. Express the relationship between 'a number is divisible by 3' and 'sum of the digits is a multiple of 3' using 'If-then' sentences.

- **Proposition:** If a number is divisible (can be divided exactly, leaving no remainder) by 3, then the sum of its digits is a multiple of 3.
- **Converse:** If the sum of the digits of a number is a multiple of 3, then the number is divisible by 3.
- **Both are true.** For example, 572 = 5 × 99 + 7 × 9 + (5 + 7 + 2). The first two parts are multiples of 3, so the number and its digit sum leave the same remainder on division by 3.
- So, a number is divisible by 3 **if and only if** the sum of its digits is a multiple of 3. ('If and only if' is used when a proposition and its converse are both true.)

Counterexamples

✓ **Concept:**

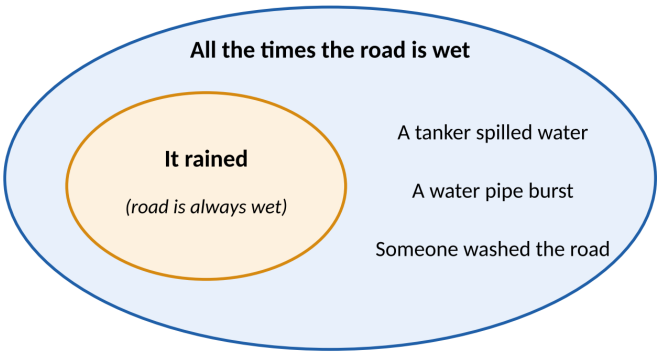
- A **counterexample** is a situation that shows why a proposition is **not necessarily true**. It **contradicts** (goes against) the proposition: the 'if' part holds, but the 'then' part does not.
- Finding a counterexample is an important part of mathematics, as it shows in a **short and convincing**

(that makes others believe it) way that a proposition is false.

- Even **one** counterexample is enough to prove a proposition false.
- But no number of examples can prove a general proposition true. For that, a **proof** (an argument that works for every case) is needed.

✓ **Examples:**

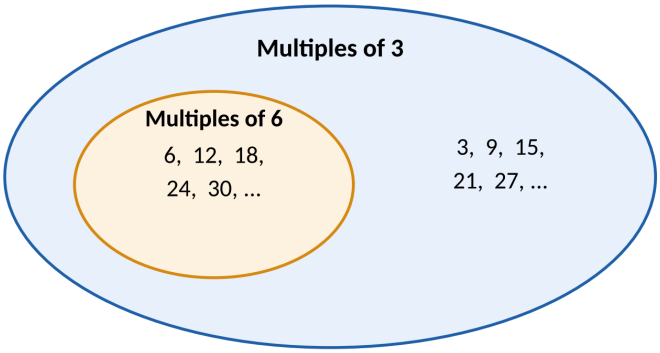
- **Example 1: Proposition:** If it rains, then the road is wet. This is **true**. **Converse:** If the road is wet, then it has rained. This **may not be true**. **Counterexample:** the road may be wet because a tanker spilled water on it, even though it did not rain.



Rain ⟹ wet road is true, but a wet road does not always mean rain.

Fig. 3: Rain always makes the road wet, but a wet road has other causes too

- **Example 2: Proposition:** If a number is a multiple of 6, then it is a multiple of 3. This is **true**, as 6k = 3 × (2k). **Converse:** If a number is a multiple of 3, then it is a multiple of 6. This is **false**. **Counterexample:** 9 is a multiple of 3 but not of 6 (also 3, 15, 21, ...).



Every multiple of 6 is a multiple of 3, but 3, 9, 15, ... are not multiples of 6.

Fig. 4: Every multiple of 6 is a multiple of 3, but not the other way round

- **Example 3 (Fermat and Euler):** the French mathematician **Pierre de Fermat** claimed that all numbers of the form $2^{(2^n)} + 1$ are prime (a number that has exactly two factors, 1 and itself). They are prime for n = 0, 1, 2, 3, 4 (3, 5, 17, 257, 65537). But the Swiss mathematician **Leonhard Euler** showed that for n = 5, $2^{32} + 1 = 4294967297 = 641 \times 6700417$, which is **composite** (a number that has more than two factors). This one counterexample **disproved** (showed to be false) Fermat's claim.

✓ **Questions:**

Q1. Identify real-life examples in which a proposition is true but not its converse. Give counterexamples.

➢ 'If it is Sunday, then the school is closed' is true. Converse: 'If the school is closed, then it is Sunday' is

false. Counterexample: the school is closed on Diwali, which may not be a Sunday.

- 'If a person lives in Mumbai, then the person lives in India' is true. Converse: 'If a person lives in India, then the person lives in Mumbai' is false. Counterexample: a person living in Delhi.
- 'If an animal is a cow, then it has four legs' is true. Converse: 'If an animal has four legs, then it is a cow' is false. Counterexample: a dog.

Q2. Give examples from geometry and number theory in which a proposition is true but not its converse, with counterexamples.

- **Geometry:** 'If a quadrilateral is a square, then its diagonals (line segments joining opposite corners of a quadrilateral) are equal' is true. The converse 'If the diagonals of a quadrilateral are equal, then it is a square' is false. Counterexample: a rectangle 4 cm by 2 cm.
- **Geometry:** 'If a triangle is equilateral, then it is isosceles' is true. The converse is false. Counterexample: a triangle with sides 5 cm, 5 cm and 3 cm.
- **Number theory:** 'If a number is a multiple of 4, then it is even' is true. The converse 'If a number is even, then it is a multiple of 4' is false. Counterexample: 6.
- **Number theory:** 'If a prime number is greater than 2, then it is odd' is true. The converse 'If a number greater than 2 is odd, then it is prime' is false. Counterexample: $9 = 3 \times 3$.

Q3. There are no known 'neat' expressions that generate only primes. Find counterexamples to: (i) All numbers of the form $4n^2 + 1$ are prime. (ii) All numbers of the form $n^2 + n + 11$ are prime. (iii) All numbers of the form $4^n + 3$ are prime.

- **(i)** n = 1, 2, 3 give 5, 17, 37 (prime). But **n = 4** gives $65 = 5 \times 13$ (not prime).
- **(ii)** n = 1 to 9 give 13, 17, 23, 31, 41, 53, 67, 83, 101 (prime). But **n = 10** gives $121 = 11 \times 11$ (not prime).
- **(iii)** n = 1, 2, 3 give 7, 19, 67 (prime). But **n = 4** gives $259 = 7 \times 37$ (not prime).

Q4. Find counterexamples to: (i) If n is a prime number, then $2^n - 1$ is a prime number. (ii) If n is an even number, then $2^n + 1$ is a prime number.

- **(i)** n = 2, 3, 5, 7 give 3, 7, 31, 127 (prime). But **n = 11** gives $2^{11} - 1 = 2047 = 23 \times 89$ (not prime).
- **(ii)** n = 2, 4 give 5, 17 (prime). But **n = 6** gives $2^6 + 1 = 65 = 5 \times 13$ (not prime).

Checking a Proposition and its Converse

✓ **Concept:**

- To check a proposition and its converse, follow these steps:
  1) Write the proposition P in the 'If X, then Y' form and find X and Y.
  2) Write the converse Q, 'If Y, then X'.
  3) Check P and Q **separately**. For a true statement, give a **proof** that works for every case. For a false statement, give a **counterexample**.
- When P and Q are both true, X and Y **imply each other**, and we say 'X **if and only if** Y'.
- **Note:** a proof must start from the 'if' part and reach the 'then' part. An argument that starts from Y and reaches X proves only the converse, not the proposition.

✓ **Examples:**

- **Example 1: perfect squares and the number of factors (both true):**
  - **P:** If n is a perfect square, then it has an odd number of factors. **Q:** If n has an odd number of factors, then it is a perfect square. (n is a positive integer.)

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- Each factor of a number has a **partner factor**, so that their product is the number. Example: for 35, 5 × 7 = 35, so 5 and 7 are partners.
- A factor-partner pair adds two factors, but a pair in which the **same number repeats**, like (6, 6) for 36, adds only one factor.

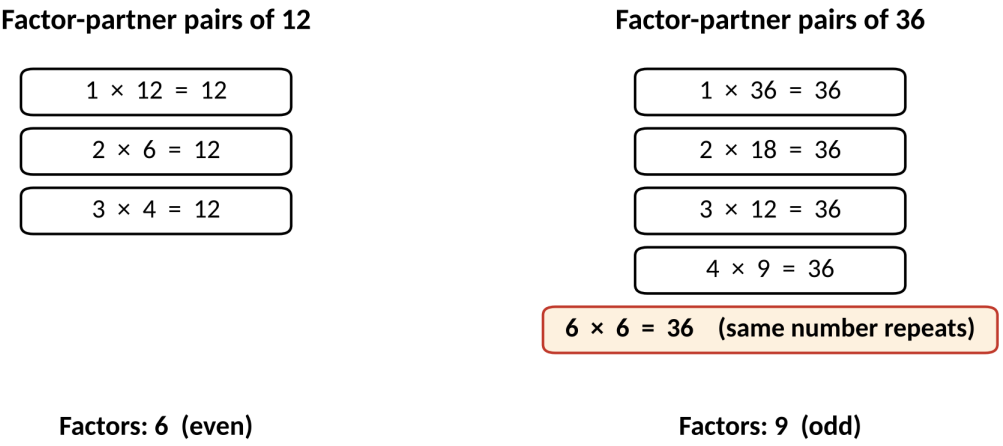


Fig. 5: Factor-partner pairs of 12 (not a perfect square) and 36 (a perfect square)

Proof of Q: a number with an odd number of factors is a perfect square:

- **Given:** n has an odd number of factors. **To prove:** n is a perfect square.

1) **Make factor-partner pairs.** For every factor f of n, its partner is n ÷ f, as f × (n ÷ f) = n. So all the factors of n can be written in pairs.
2) **Count the factors pair by pair.** A pair of two **different** numbers, like (2, 18) for 36, gives **2** factors. A pair in which the **same** number repeats, like (6, 6), gives only **1** factor, as 6 is counted once.
3) **Suppose no pair has a repeating number.** Then every pair gives 2 factors, and the total is 2 + 2 + 2 + …, which is always **even**.
4) **But the total is odd (given).** So our supposition is wrong: at least one pair must have the **same number repeating**, say (f, f).
5) **Conclusion.** For this pair, f × f = n. So n = f^2, and n is a **perfect square**.

- The same steps can be written as a chain, where each step implies the next:

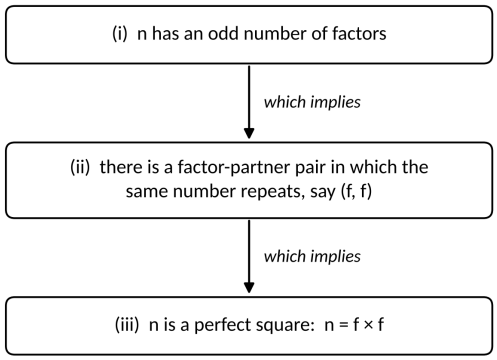


Fig. 6: The chain of steps that proves Q

- **Check with numbers:**

Number	Factor-partner pairs	Number of factors	Perfect square?
12	(1, 12), (2, 6), (3, 4)	6 (even)	No

Number	Factor-partner pairs	Number of factors	Perfect square?
16	(1, 16), (2, 8), **(4, 4)**	5 (odd)	Yes, 16 = 4 × 4
18	(1, 18), (2, 9), (3, 6)	6 (even)	No
36	(1, 36), (2, 18), (3, 12), (4, 9), **(6, 6)**	9 (odd)	Yes, 36 = 6 × 6
49	(1, 49), **(7, 7)**	3 (odd)	Yes, 49 = 7 × 7

- Whenever the number of factors is odd, a pair like (4, 4), (6, 6) or (7, 7) is present, and that repeating number is the square root.
- **Note:** this argument starts from 'odd number of factors' and ends at 'perfect square'. So it proves **only Q**, not P.

Proof of P: a perfect square has an odd number of factors:

- **Given:** n is a perfect square, n = f × f. **To prove:** n has an odd number of factors.

1) (f, f) is a factor-partner pair in which the same number repeats. It gives 1 factor.
2) n cannot have a second such pair (g, g). If g × g = f × f with g bigger than f, then g × g would be bigger than f × f, which is impossible. So g = f.
3) So every other pair is made of two different numbers and gives 2 factors.
4) Total = 1 + 2 + 2 + … , which is **odd**.

- So both P and Q are true: n is a perfect square **if and only if** it has an odd number of factors.

Example 2: area and congruence of triangles (P false, Q true):

Same area, but different shapes: NOT congruent

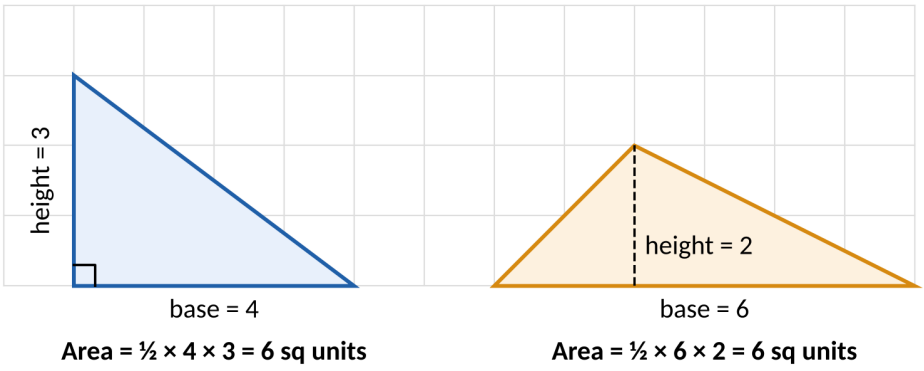


Fig. 7: Two triangles with the same area that are not congruent

- **P:** If two triangles have the same area, then they are congruent. **False.** Counterexample: a right triangle with base 4 and height 3, and a triangle with base 6 and height 2, both have area 6 sq units but are not congruent.
- **Q:** If two triangles are congruent, then they have the same area. **True**, as congruent triangles are exactly the same in shape and size, so they cover the same region.
- **Note:** 'same area' only means that the two triangles cover the same amount of surface, while 'congruent' means exactly the same shape and size. Congruence (being exactly the same in shape and size) is the stronger condition: it always gives equal area, but equal area does not give congruence.

Example 3: converse of the Baudhāyana–Pythagoras theorem (both true):

- **Theorem:** Let a, b, c be the sidelengths of a triangle. If the triangle is right-angled, then $a^2 + b^2 = c^2$.

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- **Converse:** Let a, b, c be the sidelengths of a triangle. If $a^2 + b^2 = c^2$, then the triangle is right-angled.
- **Baudhāyana** was an ancient Indian mathematician (about 800 BCE). His **Baudhāyana Śulbasūtra**, a text on making fire altars, states this result for the diagonal of a rectangle, long before Pythagoras.
- **Proof of the converse:** let ΔABC have BC = a, CA = b, AB = c with $a^2 + b^2 = c^2$. Construct a right triangle ΔXYZ, right-angled at Z, with YZ = a and XZ = b.
- By the theorem in ΔXYZ, $XY^2 = a^2 + b^2 = c^2$, so **XY = c**.
- Now BC = YZ, CA = ZX and AB = XY, so **ΔABC ≅ ΔXYZ** (SSS rule). Hence ∠C = ∠Z = 90°, and ΔABC is right-angled.
- **Example:** sides 3, 4, 5 form a right triangle, as $3^2 + 4^2 = 25 = 5^2$. Sides 4, 5, 6 do not, as $4^2 + 5^2 = 41 \neq 36$.

(a) Triangle ABC with $a^2 + b^2 = c^2$ (b) Right triangle XYZ with YZ = a, XZ = b, ∠Z = 90°

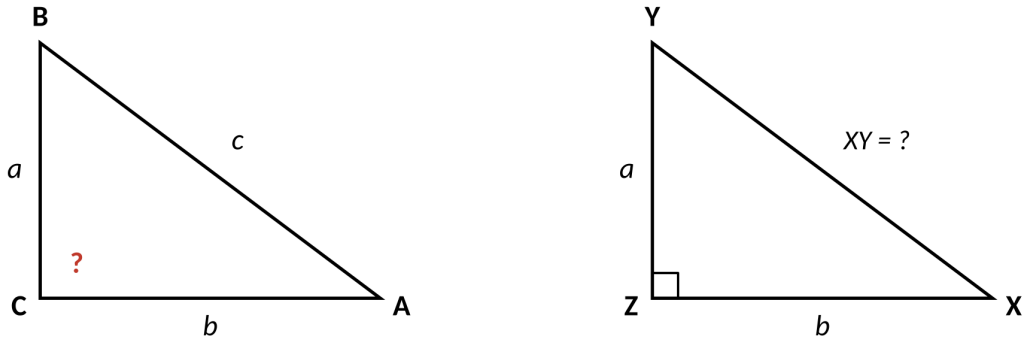
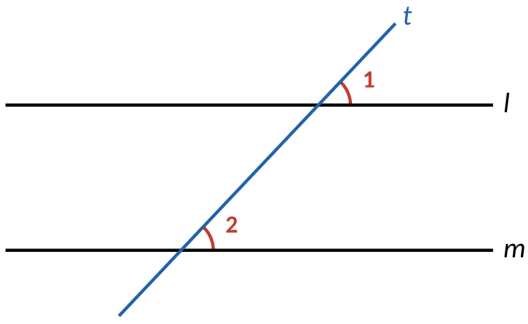


Fig. 8: Triangle ABC with $a^2 + b^2 = c^2$, and the right triangle XYZ with sides a and b

✓ **Questions:**

Frame the converse of each proposition in Q1–Q12. Check whether each statement is true. Justify true statements and give a counterexample for false ones.

Q1. If two lines are parallel, then the corresponding angles formed by a transversal are equal.



∠1 and ∠2 are corresponding angles

Fig. 9: Lines l and m cut by the transversal t

➢ **Converse:** If the corresponding angles (angles in the same position at the two crossing points made by a transversal) formed by a transversal (a line that cuts two or more lines) on two lines are equal, then the lines are parallel.

➢ **Proposition: True** (corresponding angles property of parallel lines).

➢ **Converse: True.** If the corresponding angles are equal, both lines make the same angle with the transversal, so they never meet; hence they are parallel.

Q2. If a quadrilateral is a square, then all its angles are equal.

➢ **Converse:** If all the angles of a quadrilateral are equal, then it is a square.

➢ **Proposition: True.** All angles of a square are 90°.

➢ **Converse: False.** Equal angles means each is 360° ÷ 4 = 90°, which gives a rectangle. Counterexample: a rectangle 4 cm by 2 cm.

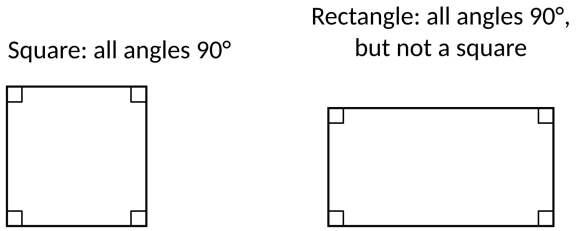
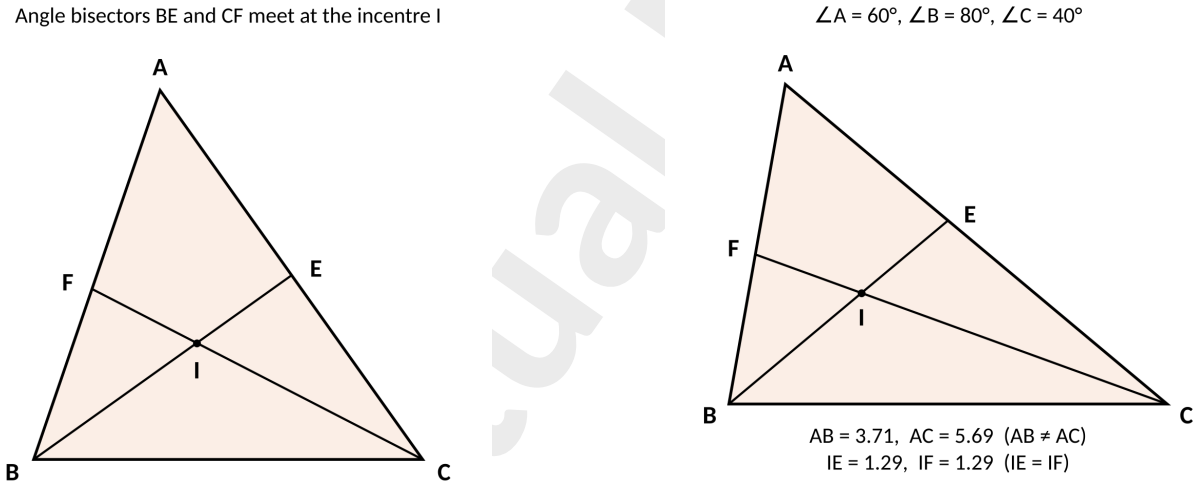


Fig. 10: A square and a rectangle both have all angles equal to 90°

Q3. In ΔABC, the bisectors of the angles at B and C meet at the incentre I, and when extended they meet the opposite sides at E and F respectively. Proposition: If AB = AC, then IE = IF.



(a) Bisectors BE and CF meet at the incentre I *(b) Counterexample: ∠A = 60°, IE = IF but AB ≠ AC*

Fig. 11: Incentre figure for Q3 and the counterexample to its converse

➢ **Converse:** If IE = IF, then AB = AC.

➢ **Proposition: True.** AB = AC gives ∠B = ∠C, so ∠IBC = ∠ICB and IB = IC. In ΔFBC and ΔECB: BC is common, ∠FBC = ∠ECB and ∠FCB = ∠EBC, so ΔFBC ≅ ΔECB (ASA) and CF = BE. Hence IE = BE − IB = CF − IC = IF.

➢ **Converse: False.** Counterexample: ∠A = 60°, ∠B = 80°, ∠C = 40°. Here AB ≠ AC (AB ≈ 3.71, AC ≈ 5.69), but IE = IF ≈ 1.29.

➢ This happens as, when ∠A = 60°, ∠FIE = 90° + ½∠A = 120°, so A, F, I, E lie on a circle. AI cuts ∠A into two equal angles, and equal angles at A in this circle cut off equal chords IE and IF.

Q4. If x = y, then a + x = a + y, where x, y and a are any three numbers.

➢ **Converse:** If a + x = a + y, then x = y.

➢ **Both true.** Adding a to equal numbers gives equal results; subtracting a from both sides of a + x = a + y

gives x = y. Both are used while solving equations.

Q5. If a and b are perfect squares, then ab is a perfect square.

- **Converse:** If ab is a perfect square, then a and b are perfect squares.
- **Proposition: True.** $a = m^2$, $b = n^2$ gives $ab = (mn)^2$.
- **Converse: False.** Counterexample: a = 2, b = 8, $ab = 16 = 4^2$, but 2 and 8 are not perfect squares.

Q6. If x = y, then $x^2 = y^2$ (x, y real numbers).

- **Converse:** If $x^2 = y^2$, then x = y.
- **Proposition: True.** Equal numbers have equal squares.
- **Converse: False.** Counterexample: x = 3, y = −3 give $x^2 = y^2 = 9$, but $x \neq y$.

Q7. If x = y, then $x^3 = y^3$ (x, y real numbers).

- **Converse:** If $x^3 = y^3$, then x = y.
- **Proposition: True.** Equal numbers have equal cubes.
- **Converse: True.** A cube keeps the sign of the number ($(-3)^3 = -27$) and a bigger number has a bigger cube, so two different numbers never have the same cube.

Q8. If n is divisible by 24, then it is divisible by both 4 and 6.

- **Converse:** If n is divisible by both 4 and 6, then it is divisible by 24.
- **Proposition: True.** $n = 24k = 4 \times (6k) = 6 \times (4k)$.
- **Converse: False.** Counterexample: 12. (4 and 6 have the common factor 2, so their LCM is 12, not 24.)

Q9. If n is divisible by 60, then it is divisible by both 5 and 12.

- **Converse:** If n is divisible by both 5 and 12, then it is divisible by 60.
- **Proposition: True.** $n = 60k = 5 \times (12k) = 12 \times (5k)$.
- **Converse: True.** 5 and 12 have no common factor other than 1. If n = 12k and 5 divides n, then 5 must divide k, so n = 60m.

Q10. If n is the square of a prime number, then it has exactly 3 factors.

- **Converse:** If n has exactly 3 factors, then it is the square of a prime number.
- **Proposition: True.** The factors of p^2 are only 1, p and p^2.
- **Converse: True.** 3 is odd, so n is a perfect square, $n = m^2$ (Example 1), with factors 1, m, m^2. If m were not prime, a factor of m between 1 and m would be a fourth factor. So m is prime.

Q11. If n is a product of two unequal prime numbers, then it has exactly 4 divisors.

- **Converse:** If n has exactly 4 divisors (numbers that divide a given number exactly), then it is a product of two unequal primes.
- **Proposition: True.** The divisors of pq are 1, p, q and pq.
- **Converse: False.** Counterexample: 8 has divisors 1, 2, 4, 8, but $8 = 2 \times 2 \times 2$.

Q12. If n and n + 3 have no factors in common (other than 1), then n is not a multiple of 3.

- **Converse:** If n is not a multiple of 3, then n and n + 3 have no factors in common.
- **Proposition: True.** If n were a multiple of 3, then n + 3 would also be one, and 3 would be a common factor.
- **Converse: True.** A common factor of n and n + 3 also divides their difference, 3, so it is 1 or 3. As n is not

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a multiple of 3, it must be 1.

Q13. 'If a number is divisible by 8, then it is divisible by both 2 and 4.' (i) Justify it. (ii) To check divisibility by 8, is it enough to check divisibility by 2 and 4?

- **(i)** $8k = 2 \times (4k) = 4 \times (2k)$, so the number is divisible by 2 and 4.
- **(ii) No.** That would mean using the converse, which is false. Counterexample: 12 is divisible by 2 and 4 but not by 8. The correct test: the number formed by the **last three digits** must be divisible by 8.

Q14. A category Q of quadrilaterals is to be made using two sticks as diagonals, joining their endpoints. (i) If Q satisfies: 'If a quadrilateral is of type Q, then it has equal diagonals', (a) should the sticks be of equal length? (b) Does it matter how they are put together? (ii) What if Q satisfies: 'If a quadrilateral has equal diagonals, then it is of type Q'?

(a) Equal sticks, crossing at the middle: rectangle

(b) Equal sticks, crossing at the middle at 90°: square

(c) Equal sticks, crossing away from the middle

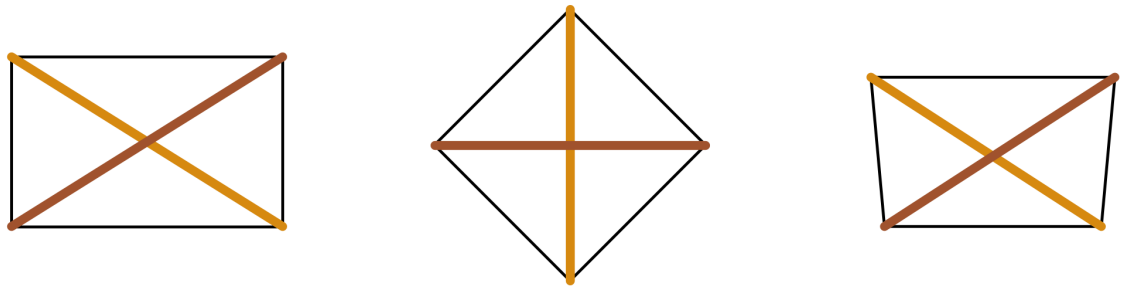


Fig. 12: Equal sticks put together in different ways give different quadrilaterals

- **(i)(a) Yes.** Every type-Q quadrilateral has equal diagonals, so unequal sticks can never give type Q.
- **(i)(b) Yes, it may matter.** Only 'type Q ⇒ equal diagonals' is given, not its converse, so equal sticks alone do not promise type Q. For example, for rectangles the sticks must cross at their middle points.
- **(ii)(a)** Equal sticks are **enough**, as every quadrilateral with equal diagonals is of type Q. But they may not be necessary, as some type-Q quadrilateral could have unequal diagonals.
- **(ii)(b) No**, it does not matter, as long as the equal sticks cross each other; every such quadrilateral has equal diagonals and so is of type Q.

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